

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
C	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
$EXPER$	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
$EXPER^2$	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
$SOUTH$	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
$UNION$	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
N	716	716	1432	1432	1432

(standard errors in parentheses)

- The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

Explain any differences in assumptions between this model and the models in part (a).

- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

(a)

The OLS estimates for 1987 and 1988 are highly consistent. The numerical differences between the coefficients in column (1) and (2) are small, typically within one standard error of each other. This suggests that the structural relationship between independent variables and $\ln(WAGE)$ remained stable over these two years.

In these individual-year OLS estimations, we are assuming intercept homogeneity. Specifically, we assume that there is no unobserved individual-specific effect correlated with the explanatory variables. By using OLS, we treat the data as a single cross-section for each year, assuming that the population parameters (including the intercept) are identical across all individuals in the sample.

(b)

According to the model, the error term is decomposed into two components. One is the individual specific effect(ui), which represents unobserved heterogeneity that is time-invariant. The other is the idiosyncratic error(eit), which varies across both individuals and time.

The key difference in assumptions is: Unlike the OLS models in part (a) which lump everything into a single error term, this panel model acknowledges that ui exists and stays constant for each individual, allowing us to control for person-specific traits that OLS might ignore.

(c)

Apart from the intercepts, the coefficient for *EXPER* shows the most significant difference. In the OLS estimations (Columns 1 and 2), the effect of experience is approximately 0.127, whereas in the Fixed Effects (FE) model (Column 3), it drops substantially to 0.0575.

The coefficient for *UNION* also shows a notable decrease from 0.144 to 0.0822. These differences suggest that the OLS estimates were biased upwards because they captured unobserved individual heterogeneity (such as ability or motivation) that is positively correlated with experience and union status. The FE model controls for these time-invariant traits, providing a more accurate estimate of the ceteris paribus effect.

(d)

$$N = 716, T = 2, K_s = 4$$

Suppose the F -statistic value is 11.68 and null hypothesis is considered as true, at 1% level of the significance, the degree of freedom is $N - 1 = 716 - 1 = 715$. Denominator degree of freedom is $NT - N - K_s = 1432 - 716 - 4 = 712$. $F_{0.99, 715, 712} = 1.190496$. The critical value at the 1% significance level is 1.190496. Reject region is $F > F_{0.99, 715, 712} = 1.190496$, $F = 11.68$, reject null hypothesis.

This indicates that there is significant unobserved individual heterogeneity among the individuals in the sample. Therefore, using a model that accounts for these individual differences is more appropriate than Pooled OLS.

(e)

- Different Assumptions:

1. Column (3): Assumes idiosyncratic errors (eit) are i.i.d., meaning there is no correlation between an individual's errors across different years.
2. Column (4): Uses Cluster-robust standard errors, which allow for correlation within the same individual's errors over time (serial correlation), which is more realistic for panel data.

- Comparison of Standard Errors:

1. Substantial Difference: The standard error for *SOUTH* shows the most significant change, nearly doubling from 0.1258 in column (3) to 0.2495 in column (4).
2. Larger or Smaller: The robust standard errors are generally larger.

The increase in standard errors indicates that ignoring the correlation within individuals leads to an underestimation of the true variability. This is particularly evident for time-persistent variables like *SOUTH*, where the traditional OLS/FE standard errors overstate the precision of the estimate.

(f)

- Coefficient Comparison: Apart from the intercept, the *EXPER* coefficient shows the most substantial difference, changing from 0.0575 in the FE model (3) to 0.0986 in the RE model (5).
- Hausman Test: We compare RE with column (3) because the standard Hausman test assumes the idiosyncratic errors are i.i.d., which matches the assumptions used in the traditional FE model.
- Based on the significant shift in coefficients (especially for *EXPER*), the Hausman test would likely reject the null hypothesis. This suggests that the unobserved individual heterogeneity (u_i) is correlated with the explanatory variables. Therefore, Random Effects is not appropriate for this model, and the Fixed Effects (FE) estimator should be preferred as it remains consistent.

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [*Hint*: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- Using the t -test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

(a)

- We estimate that the average reading score will be 5.8 points higher if the students are in a small class instead of a regular sized class. The coefficient is significantly different from zero using a 1% level of significance.
- Having a teacher aide makes no significant improvement on the average reading score relative to the regular sized class.
- Each year of teaching experience is estimated to increase average reading score by 0.49 points.
- Boys are estimated to have 6.16 point lower reading scores than girls.
- White or Asian students are estimated to have reading scores 3.91 points higher than black students.
- Those students who receive a free lunch are predicted to have an average reading score that is 14.77 points lower than those who do not.

Pooling Model

Call:

```
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +  
    freelunch, data = star.pd, model = "pooling")
```

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-107.2197	-20.2140	-3.9345	14.3391	185.9556

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	437.764253	1.346221	325.1800	< 2.2e-16 ***
small	5.822816	0.989333	5.8856	4.190e-09 ***
aide	0.817837	0.952993	0.8582	0.3908
tchexper	0.492469	0.069555	7.0803	1.611e-12 ***
boy	-6.156421	0.796128	-7.7330	1.232e-14 ***
white_asian	3.905809	0.953607	4.0958	4.264e-05 ***
freelunch	-14.771337	0.890248	-16.5924	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(b)

- The effect of being in a small class is estimated to increase average reading score by 6.49 points, which is slightly larger than the OLS estimate.
- The estimated effect of teaching experience on average reading score falls to 0.29 points per additional year of experience.

- The estimated difference between boys and girls average reading scores is slightly smaller than the OLS estimates.
- The estimated difference in average reading scores between white or Asian students and black students roughly doubles to 8 points.

Compare:

Coefficients	Estimate(OLS)	Estimate(FE)
small	5.80 ***	6.49 ***
aide	0.82	0.996
tchexper	0.49 ***	0.29 ***
boy	-6.16 ***	-5.46 ***
white_asian	3.91 ***	8.03 ***
freelunch	-14.77 ***	-14.59 ***

Oneway (individual) effect Within Model

Call:

```
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
     freelunch, data = star.pd, model = "within")
```

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-102.6381	-16.7834	-2.8473	12.7591	198.4169

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)	
small	6.490231	0.912962	7.1090	1.313e-12	***
aide	0.996087	0.881693	1.1297	0.2586	
tchexper	0.285567	0.070845	4.0309	5.629e-05	***
boy	-5.455941	0.727589	-7.4987	7.440e-14	***
white_asian	8.028019	1.535656	5.2277	1.777e-07	***
freelunch	-14.593572	0.880006	-16.5835	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(c)

The F-test statistic is 16.698.

- The rejection region is $F: F > F_{0.95, 78, 5681} = 1.27979$ so that we reject the null hypothesis that there are no significant differences between schools and conclude that there are individual differences in the school fixed effects.
- If the school indicator variables are uncorrelated with the included explanatory variables, then they should have little effect on the regression estimates.

(d)

The random-effects estimates are very similar to those from the OLS and fixed-effects models. The coefficient in small remains positive and significant, while aide remains insignificant. Other variables keep the same signs and significance, with only minor changes in magnitude.

Some variables such as white_asian, freelunch, and possibly tchexper may be correlated with school specific effects, since schools differ in socioeconomic composition and teacher quality. In contrast, small is less likely to be correlated with school effects due to random assignments within schools.

The Breusch–Pagan LM test strongly rejects the null of no random effects ($p < 0.01$), indicating that random effects are present and the RE model is preferred over pooled OLS.

(e)

Using the t-test in equation (15.36), we compare the fixed effects and random effects estimates. For *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*, the differences between FE and RE estimates are very small and not statistically significant at the 5% level. This suggests that the random effects estimator is consistent and that school-specific effects are not strongly correlated with these regressors.

```
> cat("T-statistic for BOY:", t_value_boy, "\n")
T-statistic for BOY: 0.05455779
> cat("P-value for BOY:", p_value_boy, "\n")
P-value for BOY: 0.9564927
```

For the variable BOY, the same approach was applied, but its p-value does not reject the null hypothesis, indicating no difference between the effects, and the random effects model is valid.

(f)

The Mundlak approach includes school averages of the explanatory variables to test whether the regressors are correlated with unobserved school effects.

The results show that several school-average variables, such as *small_bar*, *tchexper_bar*, *boy_bar*, and *white_asian_bar*, are statistically significant. This indicates that the explanatory variables are correlated with school-level unobserved heterogeneity.

Therefore, the random effects assumption is violated, and the fixed effects model is more appropriate.